

Cybersecurity Project Report

Incident Response Playbook and Honeypot Implementation for Threat Detection

Abstract

Cyberattacks, against organizations are getting worse fast. This means organizations need to find ways to protect themselves from cyberattacks. Cyberattacks are a problem and organizations have to do something about them. They need to use techniques to stop cyberattacks from happening.

This project is, about finding and dealing with security problems. It also looks at what happens after something goes wrong. The project focuses on the development of an Incident. We are talking about security threats and the Incident that happens because of these security threats. The goal is to find security threats respond to security threats and analyze security threats to understand the Incident.

The Response Playbook is really important. It helps us deal with problems. We are also talking about the implementation of a Honeypot system. The Incident Response Playbook provides a plan for when bad things happen. This plan is like a guide for the Response Playbook. It tells us what to do when we have a problem, with the Response Playbook and the Honeypot system. The Response Playbook and the Honeypot system work together to help us.

We need an approach to managing security incidents. This is very important. The Honeypot is a tool because it attracts attackers. At the time the Honeypot helps us analyze the attackers and what they are doing. This is really helpful for managing security incidents. The Honeypot is a big part of that. We can use the Honeypot to learn more, about security incidents and how to manage them.

When we talk about behavior we are also talking about behavior. Behavior is important. These techniques work together to make organizations more ready which is good for behavior. They help reduce the time it takes to respond to things. That is also good, for behavior. This is all because of behavior and how these techniques help with behavior.

valuable threat intelligence.

1. Bad things can happen when computers get hacked. Cybersecurity incidents like malware attacks and phishing and data breaches can cause a lot of trouble. Cybersecurity incidents are a problem. When cybersecurity incidents happen they can be very expensive. Cybersecurity incidents, like these can really hurt a companys money.

losses and damage an organization's reputation. Without a proper response plan, handling such incidents

Things become really tough and messy. An Incident Response Playbook makes sure that the security teams follow the security teams plan when the security teams are dealing with an incident. This helps the security teams to stay on track and work together smoothly which is what the security teams need to do.

When something bad happens there are predefined steps that people take. A Honeypot is a tool that helps with monitoring of malicious activities. The Honeypot is really good at keeping an eye on things that people try to do. The predefined steps are important during an incident. The Honeypot is important, for monitoring malicious activities.

The project combines both approaches to make the cybersecurity defenses stronger. This project is really, about cybersecurity defenses. It uses two different ways to do that. The project is trying to make cybersecurity defenses by using these two approaches together. This will help the cybersecurity defenses of the project.

2. Project Objectives

Incident Response Playbook: To identify and manage security incidents, minimize damage, reduce recovery

time, and standardize response procedures.

Honeypot Implementation: To attract malicious traffic, capture malware samples, analyze attack techniques,

and improve threat intelligence.

3. The Technologies That We Used

The tools that were used in this project are Java and Python for implementing things MySQL for storing all the data and SIEM tools. The Java and Python programming languages played a role in this project. We also used MySQL for data storage and SIEM tools for security. The Java and Python tools were really useful, for implementation.

such as Splunk and ELK Stack for log analysis, SOAR tools for response automation, Honeypot tools like Honeyd

and Dionaea, network monitoring tools such as Wireshark and Tcpdump, and the Linux operating system.

4. Incident Response Playbook Flowchart

Preparation

↓

Identification

↓

Containment

↓

Eradication

↓

Recovery

↓

Things I Found Out

The Incident Response Playbook has a process that it follows. This process starts with getting ready. Ends with learning from the Incident Response Playbook. The Incident Response Playbook is really important, for this. The Incident Response Playbook helps with the process.

I have learned that when things go wrong it is good to have a plan in place. This structured flow ensures that the company can handle incidents in an effective way. The incident handling process is really important, for the company. Incident handling is what the company needs to do when something bad happens.

5. Honeypot Implementation

A Honeypot is, like a trap that looks weak on purpose so it can catch people who try to attack it. The Honeypot records all the things these attackers do to it the Honeypot is used to see what the attackers are trying to do.

captures malware payloads, and provides insights into attacker behavior without risking real production systems.

Honeypot Working Flowchart

Deploy Honeypot

↓

Attacker Interaction

↓

Log & Monitor Traffic

↓

Capture Malware



Analyze Threats



Generate Threat Intelligence

6. Skills and Knowledge Gained

This project was really helpful, for me because it taught me a lot about incident management. I learned how to do SIEM log analysis which's very important. The project also helped me understand threat intelligence and how it works. I got to learn about threat intelligence. It was very useful.

gathering, malware analysis fundamentals, and network security monitoring.

7. The project shows that Incident Response Playbooks and Honeypots can really work together. Incident Response Playbooks and Honeypots are a team. This project is an example of how Incident Response Playbooks and Honeypots can be used together to achieve something.

We need to make our cybersecurity defenses stronger. The playbook is really helpful because it makes sure we handle incidents in a way. The Honeypot is also a tool, for cybersecurity defenses. We should use the Honeypot to help strengthen our cybersecurity defenses. This will help us to stay safe from cyber attacks.

This thing gives us insights, into what the attackers are doing. The attackers and their behavior are really important to understand. Together they help us detect the attackers respond to them and learn from the attackers and their behavior.

cyber threats.